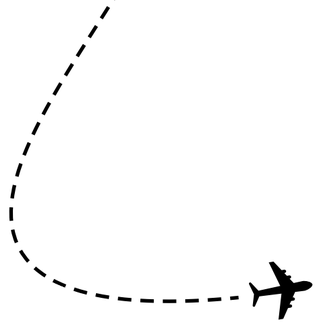
HTTP Project

Travel Agency platform with MITM simulation attack

Project by :
Sarah Yaakoubi
Rahma Abdelmoula
Montaha Bettaieb

Scholar Year: 2026/2027

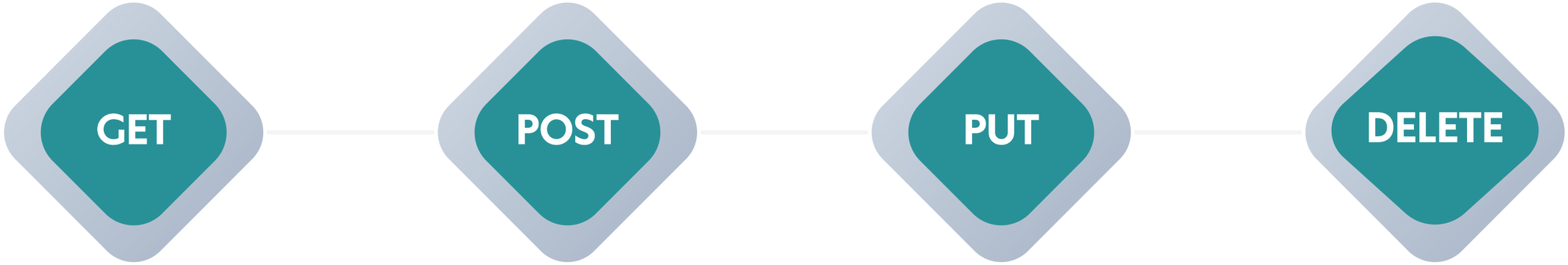
Content



- Whats HTTP
- Application presentation
- Use case senarios live demo
- Vulnerabilities
- MITM attack presenation
- MITM attack live demo
- Conclusion

What is HTTP ?

An HTTP request



An HTTP response : status codes

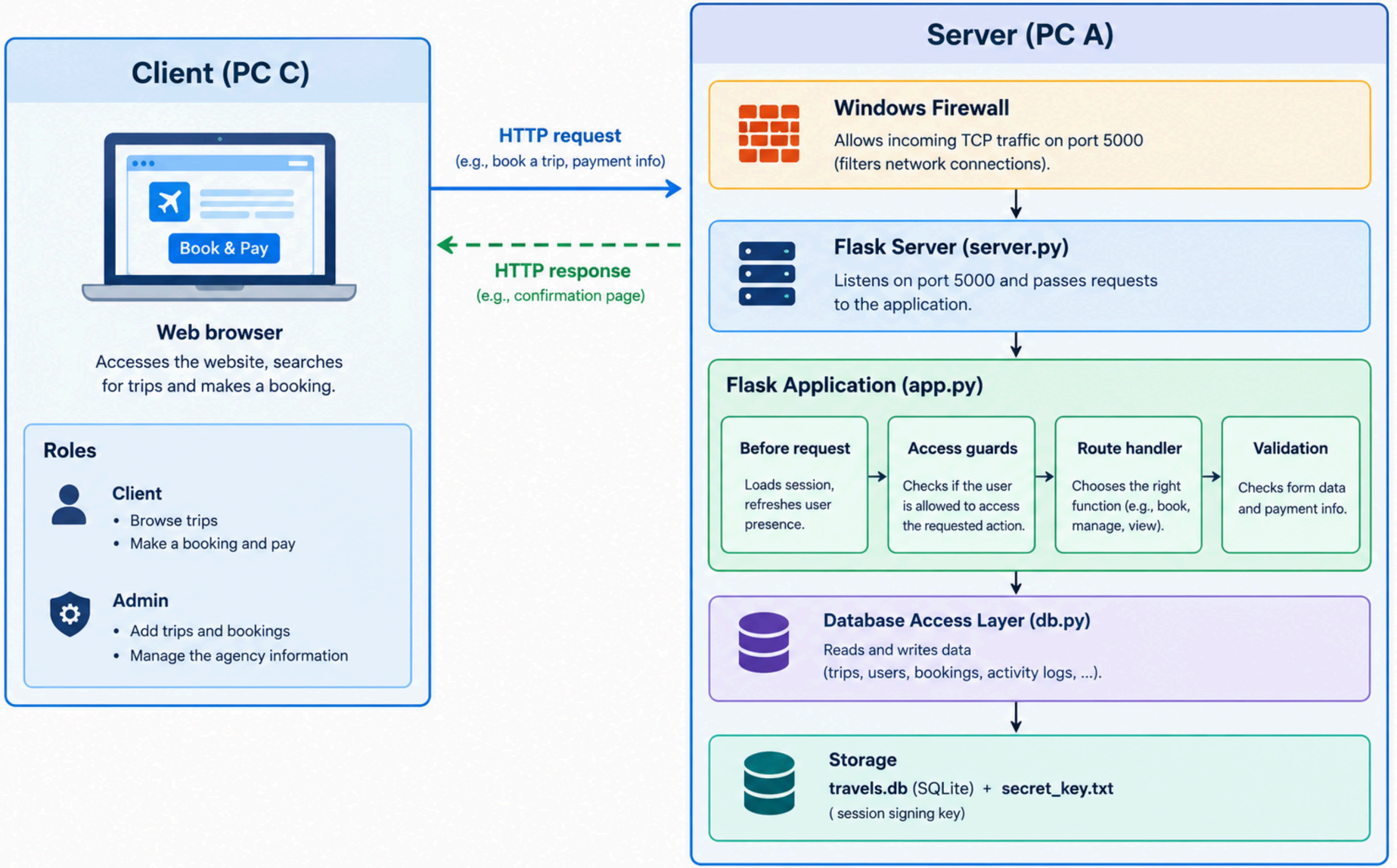
200

301

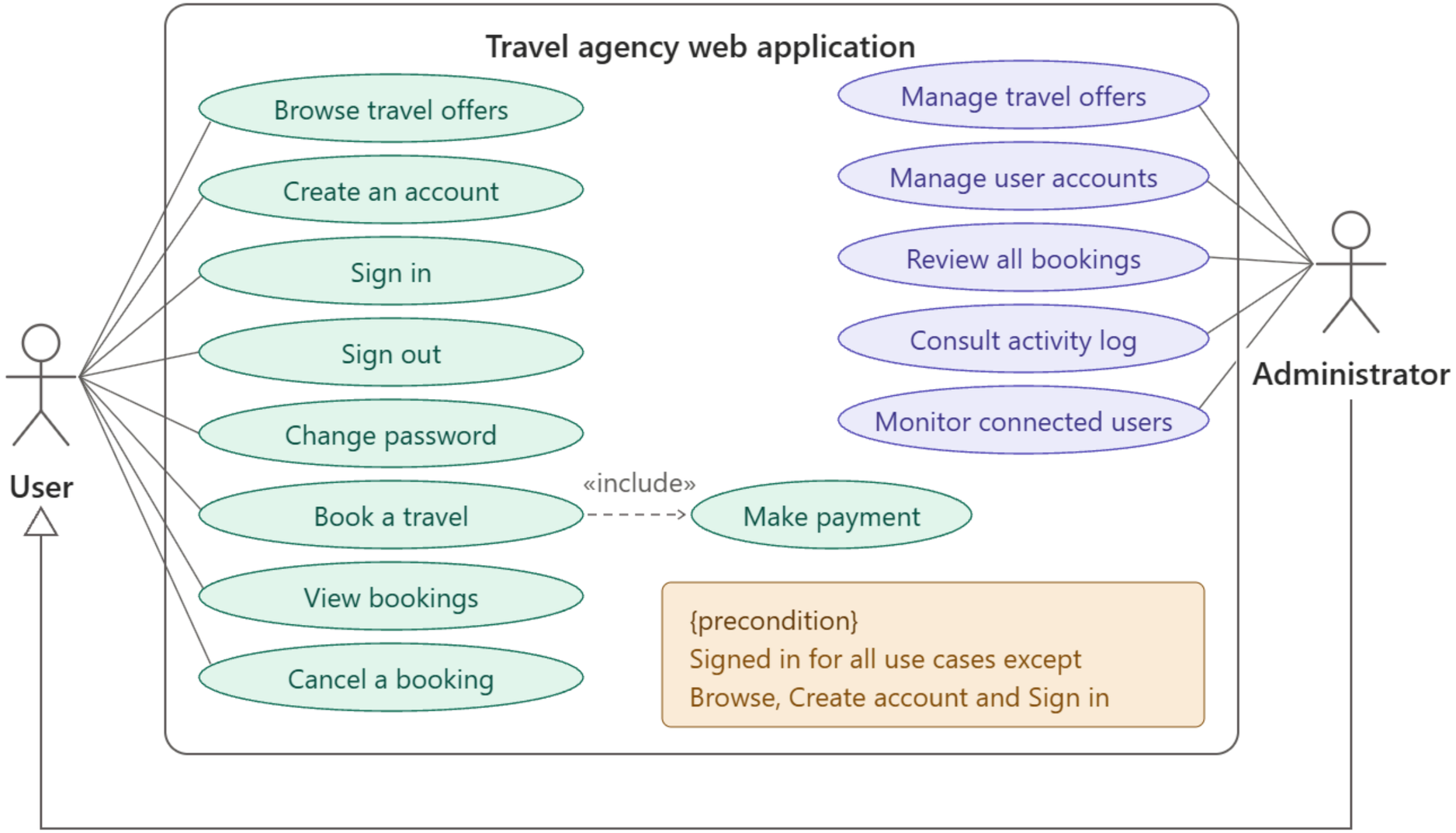
404

500

Global Architecture



Global use case



Live demo

Use case live demo

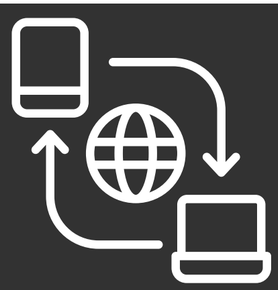


Vulnerabilities



Passwords captured in cleartext

There's no client-side hashing. POST /login and POST /signup send the password in the request body over HTTP.



No transport integrity

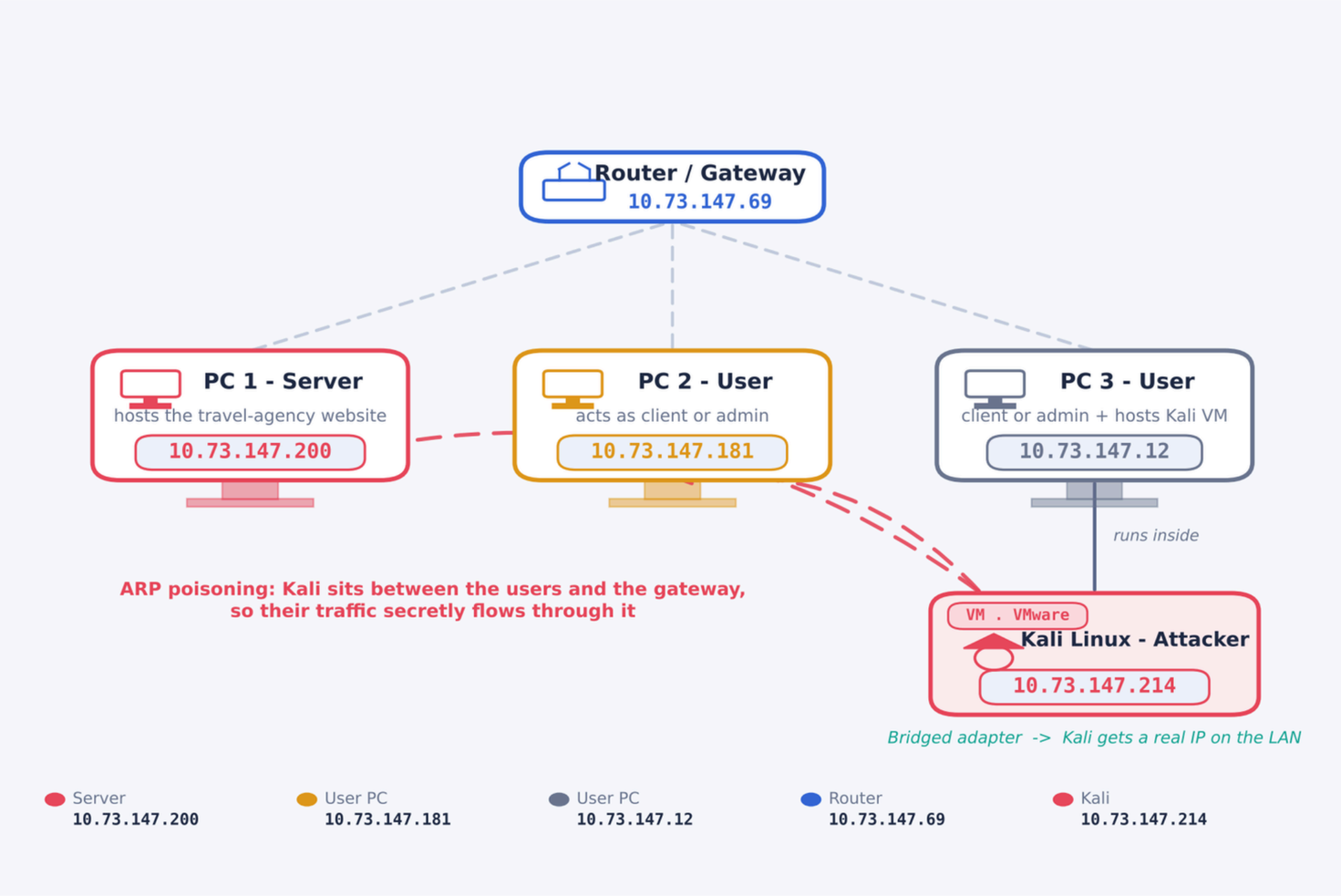
nothing proves the user is talking to the real server



Payment and ID data harvested in transit

POST /travels/<id>/book sends CIN, full address, phone, and the full card number and CVV in cleartext

Architecture



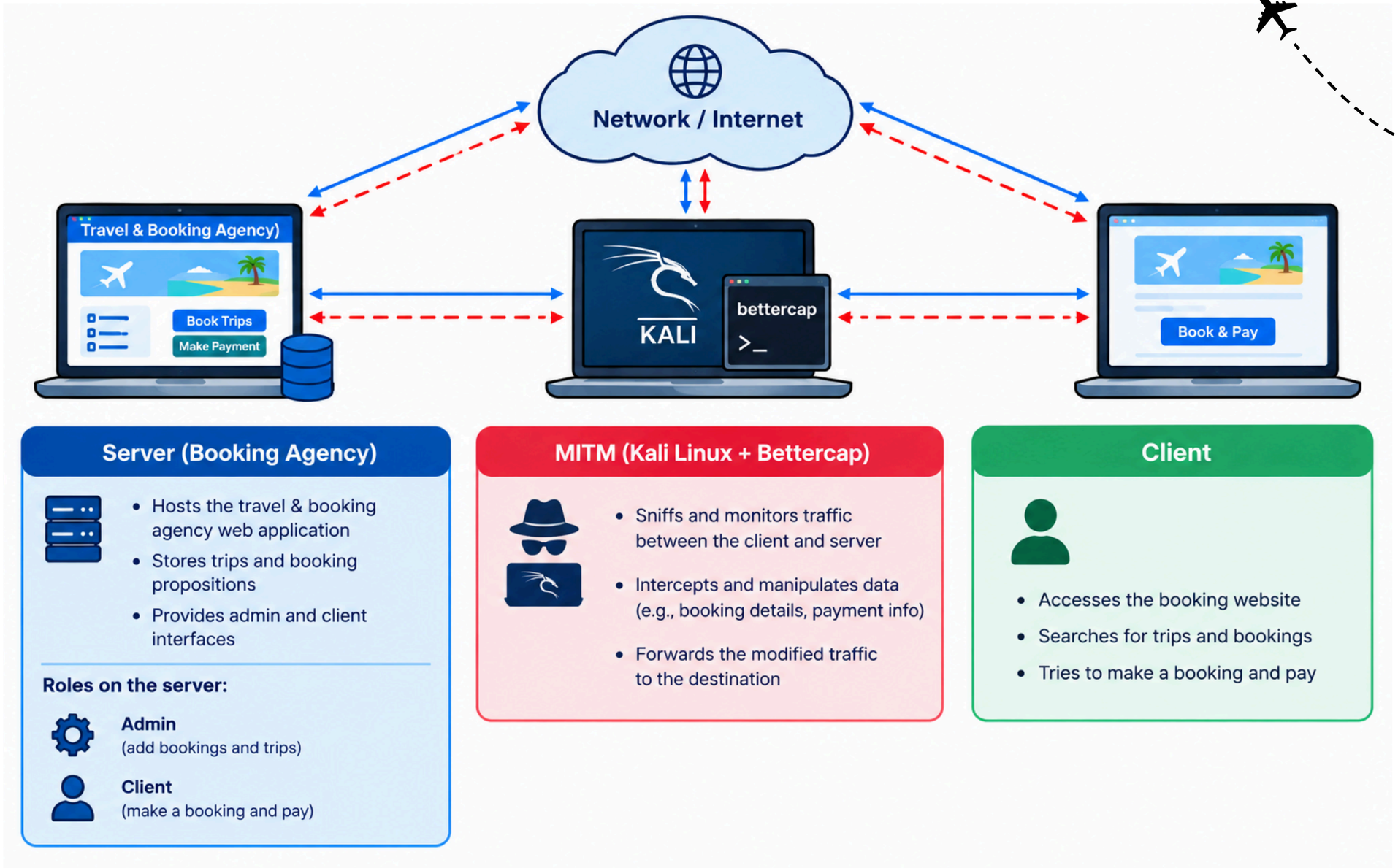
Network Configuration

Bridged : The virtual machine connects directly to the physical network and receives its own IP address from the router, functioning as an independent device that other hosts can see and communicate with.

NAT : The virtual machine shares the host's connection through an internal virtual router, allowing it to access external networks while remaining hidden from other devices on the LAN.

Host-only : The virtual machine is placed on an isolated private network limited to communication with the host machine, with no access to the external network or the internet.

MITM attack architecture



Server (Booking Agency)

- Hosts the travel & booking agency web application
- Stores trips and booking propositions
- Provides admin and client interfaces

Roles on the server:

Admin
(add bookings and trips)

Client
(make a booking and pay)

MITM (Kali Linux + Bettercap)

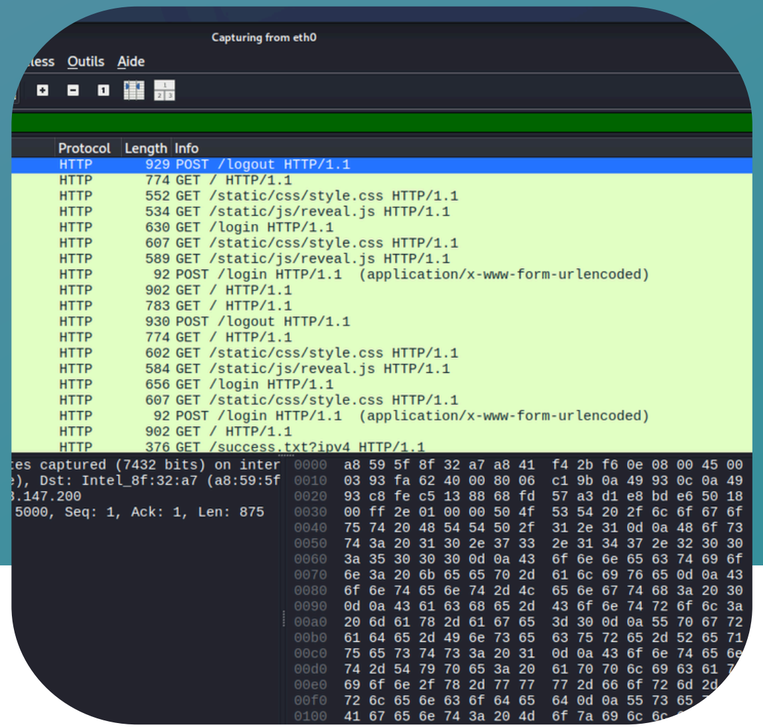
- Sniffs and monitors traffic between the client and server
- Intercepts and manipulates data (e.g., booking details, payment info)
- Forwards the modified traffic to the destination

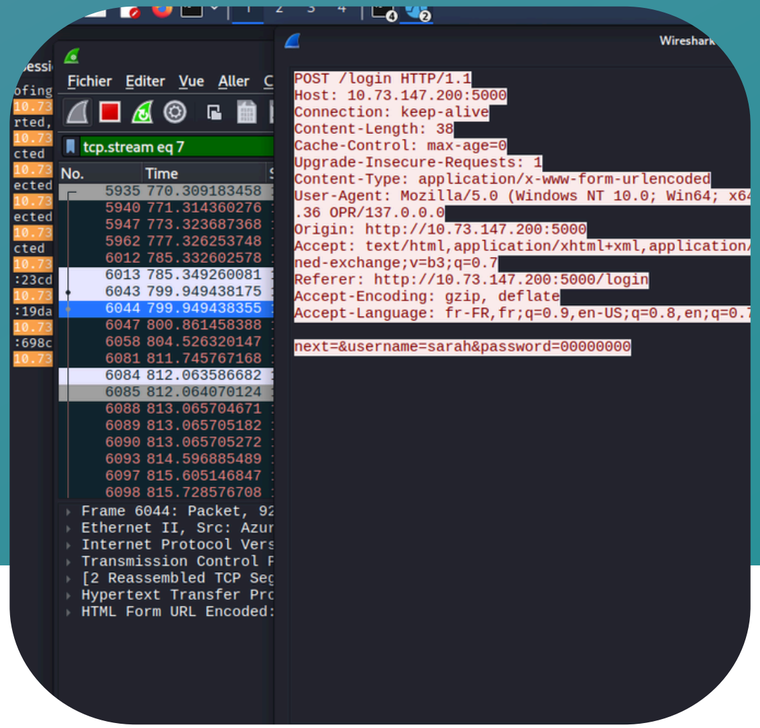
Client

- Accesses the booking website
- Searches for trips and bookings
- Tries to make a booking and pay

The 3 main steps of a man-in-the-middle attack







1. Position

2. Intercept

3. Extract

Live demo

MITM live demo



Conclusion